

Linux + Shell (EXERCISE)

Tuesday, June 10, 2025 7:31 PM

Command Exercise

Linux File System & Navigation

1. List all files (including hidden) in long format with human-readable sizes, sorted by modification time (newest first). Which flags?
2. Print the absolute path of your current directory and store it in an environment variable CUR_DIR. Which command or flags?
3. Change to your home directory using a shortcut, then in the same command change to the parent of your current directory. Which syntax?
4. Create the nested directory structure `project/{src,bin,docs}` in one command. Which flag allows this?
5. First run `echo "hello" > file.txt` and `mkdir archive`, then move `file.txt` into `archive/` renaming it to `file.bak`, prompting before overwrite. Which flag?
6. First `mkdir -p data` and `echo a > data/a.txt` and `echo b > data/b.txt`, then copy the entire `data/` directory to `backup/`, preserving timestamps and permissions. Which flags?
7. First `mkdir -p old_logs` and `echo log1 > old_logs/log1.txt`, then remove the non-empty directory `old_logs/` without any prompts. Which flags?
8. First create an empty file `notes.md`, then update its modification time to January 1 2022 at 12 PM. Which flag?
9. Clear your terminal screen. What keyboard shortcut achieves the same effect?
10. Temporarily set the environment variable `EDITOR=vim` in your current shell. Which command or syntax ensures it's inherited by child processes?
11. What command shows you the directory hierarchy from `/` down to your current directory?

Permissions & Ownership

1. Set permissions on `script.sh` so owner has `rwx`, group has `r-x`, others none—first using numeric mode, then symbolic mode. Which syntaxes?
2. Change owner to `alice` and group to `staff` for `/var/www/html` recursively. Which flag?
3. Create a symbolic link `latest.log` pointing to `/var/log/syslog`. Which flag, and how does a symbolic link differ from a hard link?
4. You have a file owned by `root:root` at `/opt/data.txt`. Change its owner to `you` and group to `users`, without affecting anything else. Which command?

Text Processing, Redirection & Piping

1. First `echo A > a.txt`, `echo B > b.txt`, and `echo C > c.txt`, then concatenate `a.txt`, `b.txt`, and `c.txt` into `all.txt` and display non-printing characters. Which flags?
2. Print the literal string `Hello, $USER!` (without expanding `$USER`), then interpret `\n` as a newline. Which flags?
3. Open a large log file with line numbers displayed using `less`. Which flag, and what key do you press to search forward for “error”?
4. Display the manual page for `grep` and jump directly to the “OPTIONS” section. How do you quit?
5. First `echo -e "a\nb\nc" > A.txt` and `echo -e "b\nc\nd" > B.txt`, then list only the lines common to `A.txt` and `B.txt`. Which flag hides lines unique to the first file?
6. First `echo -e "10\n2\n30\n4" > numbers.txt`, then sort the numbers in numeric order. Which flag? How do you reverse the sort?
7. Count the number of unique IP addresses in `/var/log/nginx/access.log`. Sketch a one-liner using pipes.
8. Redirect both stdout and stderr of `rsync` into `sync.log`, appending rather than overwriting. How?

Compression & Archiving

1. First `mkdir logs` and `echo test > logs/a.log` and `echo err > logs/b.log`, then create a gzipped archive `logs.tar.gz` of `logs/` and extract it into `restore/`. Which flags?
2. Extract archive `.zip` into `dest/` without any prompts. Which flag? How do you list its contents without extracting?
3. First touch `a.png b.png c.png`, then create a zip archive `images.zip` containing all `.png` files and prompt for a password. Which flag?
4. Create a `tar.bz2` archive of `/etc` using `bzip2` compression. Which flag?

Networking & Remote Access

1. Connect to user@remote.server.com on port 2222 via `ssh`. Which flag? How do you enable X11 forwarding?
2. Trace the network path to `example.com`, limiting max hops to 16. Which flag?
3. Recursively download only `.pdf` files from <http://example.com/files/> with `wget`. Which flags? How do you set a 30 s timeout?
4. Use `curl` to fetch <https://api.example.com/data> following redirects, and save the output to `data.json`. Which flags?

System Monitoring & Process Management

1. Show all processes in a full-format listing. Which flags? How do you filter only your own processes?
2. Send a graceful `TERM` signal to process ID 12345. Then kill all processes named `worker` with `SIGKILL`. Which signals or flags?
3. Report disk usage of all filesystems in human-readable units. Which flags? How do you skip pseudo-filesystems like `tmpfs`?
4. Mount `/dev/sdb1` on `/mnt/usb` as read-only. Which flags? How do you display all current mounts?
5. Continuously follow the end of `/var/log/syslog`. Which flag lets you start by showing the last 50 lines then continue following?
6. Show memory usage and swap usage in mebibytes. Which command and flags?
7. Display system uptime and load averages. Which command?

Package & Service Management

1. On Debian/Ubuntu, update package lists and upgrade all packages; on Arch, synchronize and update; on CentOS, install `htop`; on RHEL, query which package provides `/usr/bin/wget`. Which commands?
2. Run `systemctl restart sshd` as root via `sudo`. How do you edit `/etc/hosts` in your default editor under `sudo`?
3. Start the `nginx` service on a SysVinit system, then check its status. Which commands or flags?
4. Enable and start a service called `myapp` with `systemd`, then show its logs via `journalctl` in follow mode. Which commands?

Environment Variables

1. Echo the current value of `$HOME`. Then create your own variable `MY_VAR="DataEngineering"` and display it.
2. Make `MY_VAR` permanent by adding it to your shell startup file. Which file(s) and syntax?
3. Prepend `/opt/bin` to your `PATH` for the current session. Which command? How do you make that change permanent?

Shell Exercise

• Bulk Rename & Report

Prerequisite: Create a directory `sample_images/` containing at least five `.jpg` files with arbitrary names.

Question:

Write a Bash script `rename_and_report.sh` that takes two arguments, a directory path and a filename prefix, renames every `.jpg` in that directory to `<prefix><zero-padded-sequential-number>.jpg`, keeps count of renamed files, and at the end echoes "Renamed X files in <dir> using prefix '<prefix>.'" Include a `-n` (dry-run) flag that shows what would be renamed without actually doing it.

Prerequisite: Create a file `access.log` with at least 10 lines in Apache/Nginx common-log format, containing a mix of HTTP status codes (e.g. 200, 301, 404, 500).

Write a Bash script `summarize_logs.sh` that accepts the log file path, parses each line to extract the status code, accumulates counts per code (using an associative array or equivalent), and then prints a table of “Status Count” sorted by count descending. Add an optional `--top N` flag to display only the top N most frequent codes.

1. What is the difference between terminal, shell, and Bash?
2. Describe these Linux concepts: redirect (>), append (>>), pipe (|).
3. What are tar files and why do we use them?
4. What are environment variables and how are they used by applications?
5. How can you search for strings in a text file? Give at least two different commands.
6. How can you find the largest file in a directory (and its subdirectories)?
7. Why do we write shell scripts when we have higher-level languages like Python?
8. How do you identify processes running in the background in Linux?